INCH-POUND

MIL-F-44070C
24 April 1991
SUPERSEDING
MIL-F-44070B
3 April 1987

MILITARY SPECIFICATION

FRUITS, FREEZE-DEHYDRATED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers freeze-dehydrated blocks (compressed or formed) of sweetened fruits, in flexible bags for use as a component of the meal, ready-to-eat, individual.

1.2 Classification. The fruits shall be of the following kinds as specified (see 6.1):

Kinds:

I - Fruit Mix
II - Peaches
III - Pears
IV - Strawberries

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8915

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are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-C-1752	-	Cushioning Material, Unicellular Polyethylene Foam Flexible (For Packaging Purposes)
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular Polypropylene Foam

STANDARDS

FEDERAL

FED-STD-595	-	Colors Used in Government Procurement

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Frozen Peaches
U.S. Standards for Grades of Frozen Pineapple
U.S. Standards for Grades of Frozen Strawberries
U.S. Standards for Grades of Canned Pears
U.S. Standards for Grades of Canned Sweet Cherries
U.S. Standards for Grades of Canned Pineapple

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(Copies are available from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2077, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique
F 88 - Seal Strength of Flexible Barrier Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 N. 19th Street, Arlington, VA 22209.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

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(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Peaches (kinds I and II). The peaches shall be ripe or soft ripe, fresh or individually quick frozen, peeled and pitted of the Rio Osa, Gem, Fay-Elberta, or similar yellow freestone variety suitable for freeze-drying. The peaches shall be in the form of 1/2 ± 1/8 inch strips.

3.2.2 Pears (kind I). The pears shall be mature, fresh, canned, or individually quick frozen pears of the Bartlett variety which have been peeled, cored, and cut into 1/2 ± 1/8 inch thick strips. Canned pears, when used, shall be U.S. Grade B or better of the U.S. Standards for Grades of Canned Pears.

3.2.3 Pears (kind III). The pears shall be mature, fresh, Bartlett variety and shall have a hardness reading of not less than 2 pounds nor greater than 5 pounds on a fruit pressure tester (see 6.3). The pears shall be peeled, cored, and cut into 1/2 ± 1/8 inch thick strips.

3.2.4 Pineapple (kind I). The pineapples shall be mature, fresh, canned or individually quick frozen, peeled, and cored of the Cayenne, Queen group or similar variety suitable for freeze-drying. The pineapple shall be in the form of 5/16 to 1/2 inch thick small tidbits having a 3/8 to 3/4 inch long outside arc and an 11/16 to 1-1/4 inch length measured along the radius from the inside arc to the outside arc. Canned pineapple, when used, shall be U.S. Grade B or better of the U.S. Standards for Grades of Canned Pineapple, shall be packed in pineapple juice, and may be used without draining.

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3.2.5 Maraschino cherries (kind I). The maraschino cherries shall be red (nonbleeding) of any normal commercial variety in the form of approximate halves. The maraschino cherries shall be U.S. Grade B or better of the U.S. Standards for Grades of Canned Sweet Cherries. Alternatively, cherry halves which meet the color, odor, flavor, and texture characteristics of maraschino cherries, but which are packed in a syrup of less than 37° Brix but not less than 15° Brix may be used.

3.2.6 Strawberries (kind IV). The strawberries shall be Midway, Sure Crop, Northwest, or similar high flavor variety suitable for freeze dehydration. Fresno and Teoga may be used when they provide high flavor, are suitable for freeze dehydration, and have received prior approval from the contracting officer. The strawberries shall be whole with the caps removed and shall be of such size that not less than 95 percent by weight are 5/8 inch or more in diameter when measured at right angles to a straight line running from stem to apex. Strawberries too large to meet the sweetening ingredient addition requirements in 3.3.4 and the packing requirement may be sliced with a slicer set at 1/2 inch.

3.2.7 Sodium sulfite or sodium bisulfite. The sulfite(s) used shall be of a grade comparable to the Food Chemicals Codex.

3.2.8 Ascorbic acid. Ascorbic acid used shall be of a grade comparable to the Food Chemicals Codex.

3.2.9 Sweetening ingredients. Sweetening ingredient shall be white, refined, granulated cane or beet sugar (sucrose) or sugar (sucrose) sirup with a minimum Brix value of 28.6 degrees.

3.3 Preparation and processing.

3.3.1 Proportioning of fruits (for kind I). The fruit shall be mixed in the percentages indicated below. The percentages shall be based on the drained weight of the fruit at the time it is added to the mix prior to final freezing.

Fruit	Percent by weight	
	Minimum	Maximum
Peaches	30	50
Pears	25	45
Pineapples	6	16
Maraschino cherries	2	6

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3.3.2 Sulfiting. The fruit with the exception of kind IV and canned fruit when applicable shall be treated in a sodium sulfite or sodium bisulfite solution, or a combination of both, by dipping or spraying or in any similar manner as to result in uniform distribution. The sulfur dioxide content shall comply with 3.4.1. Fruit shall be adequately drained before unit packing.

3.3.3 Ascorbic acid addition. Ascorbic acid at a level not exceeding 300 mg per pound of the fruit, including sugar, shall be added.

3.3.4 Sweetening ingredient addition. Dry sugar (sucrose), or 28.6 degree Brix or higher sugar (sucrose) sirup shall be uniformly added prior to freezing (refreezing where applicable; see 3.3.5) at a ratio of not less than eight parts fruit to one part sugar (sucrose) by weight. Any dry sugar shall be thoroughly dissolved to prevent puncture of the flexible unit pack by undissolved sugar crystals.

3.3.5 Forming for freezing. Frozen peaches may be partially thawed and used instead of fresh for kinds I and II. Individually quick frozen fruits (for kind I) shall not be allowed to thaw before addition to the fruit mix. The prepared fruit with sugar added shall be compacted and frozen into uniform blocks with a minimum of voids. The blocks shall be of such size that they can be cut into smaller blocks of a size not to exceed 4 by 3-1/2 by 9/16 inches. The product shall be compacted by hand or by machine to produce a uniform product with a minimum of voids. The fruit shall be evenly distributed within the block. As an alternative to compaction and cutting into blocks, fruit and sugar may be molded into blocks not to exceed 4 by 3-1/2 by 9/16 inches.

3.3.6 Freezing and storage. The compacted or molded product shall be placed in a blast freezer with air temperature ranging from minus 15°F to minus 30°F or on a plate freezer, for such time as is necessary to insure complete freezing. After freezing, the product shall be stored at 0°F or lower. The frozen fruit blocks shall not be produced prior to award of the contract or purchase order.

3.3.7 Frozen fruit quality. After preparation and freezing, and prior to dehydration, the fruit components shall be of the following quality:

a. Frozen peaches shall have normal flavor and good color, be practically free from defects, and possess at least a reasonably good character for soft-ripe peaches in accordance with the U.S. Standards for Grades of Frozen Peaches.

b. Frozen pears shall have normal flavor and odor and characteristic, typical, practically uniform light yellow-white, greenish-white, or beige-white color, be practically free from defects, and possess a reasonably tender fleshy texture reasonably free from noticeable graininess or toughness.

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c. Frozen strawberries shall have good flavor and good color, be practically free from defects, and possess at least a reasonably good character in accordance with the U.S. Standards for Grades of Frozen Strawberries.

d. Frozen pineapple shall have normal flavor and odor and practically uniform bright characteristic yellow color, be practically free from defects, and possess at least reasonably good character in accordance with the U.S. Standards for Grades of Frozen Pineapple.

e. Maraschino cherry halves or alternate cherry halves in syrup shall have a uniform characteristic red maraschino cherry color and a distinct characteristic maraschino cherry flavor and odor, be practically free from defects, and possess a firm fleshy texture.

3.3.8 Cutting. Prior to dehydration, the compacted product shall be cut or sawed into blocks that meet the dimension requirements of 3.3.5. This operation shall be performed as rapidly as possible and at no time during this procedure shall the temperature of the product exceed 20°F. After cutting, the product shall not exceed minus 20°F until freeze-dehydration. Such storage period shall not exceed 72 hours. The minus 20°F storage reduces the tendency for foaming (especially strawberries) during freeze-dehydration.

3.3.9 Freeze-dehydration. The frozen product blocks shall be freeze-dehydrated at absolute pressure not exceeding 1.0 millimeter of mercury for more than 5 minutes during the first three quarters or 10 minutes during the last quarter of the drying cycle. At no time shall there be evidence of thawing in the product. After dehydration has been completed, the pressure in the freeze-drying chamber shall be equalized to atmospheric pressure with nitrogen and the product shall be unit packed immediately as specified in 5.1, or shall be held under an atmosphere of nitrogen (not more than 2 percent oxygen) not more than 16 hours until unit packing can be effected.

3.4 Finished product requirements.

3.4.1 Dehydrated product.

a. There shall be no foreign material such as, but not limited to, dirt, insect, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. There shall be no evidence of incomplete dehydration, such as damp or soggy areas.

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e. Product shall show no evidence of having thawed (as evidenced by glazed or collapsed areas) or of scorching.

f. Product shall conform to the following minimum weight requirements:

Kind	Average block weight	Individual block weight
I	13.0 grams	10.0 grams
II	12.0 grams	10.0 grams
III	16.0 grams	13.0 grams
IV	14.0 grams	10.0 grams

g. The blocks shall not be crushed.

h. Maximum dimensions of product shall be 4 by 3-1/2 by 9/16 inches.

i. Moisture content of the dehydrated product shall not exceed 2.5 percent.

j. Sulfur dioxide content (kinds I, II, and III only) shall be 650 ± 450 parts per million.

3.4.2 Rehydrated product.

a. The product shall be completely rehydrated by rehydrating in 2 fluid ounces of cold water (55° ± 5°F) for the following time:
(1) Kinds I or II - 20 minutes
(2) Kinds III or IV - 15 minutes

b. Every piece of rehydrated fruit shall show complete penetration of water to the center.

c. Product shall have characteristic color, odor, and flavor of the specified kind.

3.4.3 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, and Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5.1.2 Bag material certification. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements.

Material requirement	Requirement paragraph	Test procedure
Thickness of films	5.1.1.1 and 5.1.1.2	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Aluminum foil thickness	5.1.1.1 and 5.1.1.2	As specified in QQ-A-1876
Laminated material identification and construction	5.1.1.1 and 5.1.1.2	Laboratory evaluation
Color of laminated material	5.1.1.1 and 5.1.1.2	Visual evaluation

4.5.1.3 Unfilled preformed bag certification. A certificate of compliance may be accepted as evidence that unfilled bags conform to the requirements specified in 5.1.1.1. When deemed necessary by the USDA, testing of the unfilled preformed bags for seal strength shall be as specified in 4.5.1.3.1.

4.5.1.3.1 Unfilled preformed bag seal strength testing. The unfilled bags shall be tested for seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in bags. The sample unit shall be one unfilled bag. The sample size shall be the number of bags indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each bag in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength of each seal shall be calculated by averaging the test results of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement or any seal failing to meet the average seal strength requirement specified in 5.1.1.1 shall be classified as a major defect and shall be cause for rejection of the lot.

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4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Net weight inspection. The net weight of the dehydrated product shall be determined by weighing each sample unit on a suitable scale tared with a representative empty bag. Any individual net weight of less than 13.0 grams for kind III or 10.0 grams for kinds I, II and IV shall be classified as a minor defect. The lot size shall be expressed in bags. The results shall be reported to the nearest 0.1 gram. The sample unit shall be one filled and sealed bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample average net weight is less than 16.0 grams for kind III, 13.0 grams for kind I, 12.0 grams for kind II, or 14.0 grams for kind IV, as applicable.

4.5.4 Filled and sealed bag examination. The filled and sealed bags shall be examined for the defects listed in table I. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

TABLE I. Filled and sealed bag defects 1/

Category		Defect
Major	Minor	
101		Not clean 2/
102		Tear, hole, or open seal
103		Evidence of delamination 3/
104		Seal width less than 1/16 inch 4/
105		Bag has foreign odor
106		Presence of flex fractures in the aluminum foil 5/ 6/
	201	Required labeling missing, incorrect, illegible, or that smudges
	202	Closure or top seal extends into or below tear notch 7/

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TABLE I. Filled and sealed bag defects 1/ (cont'd)

Category		Defect
Major	Minor	
	203	Tear notch or serrations missing or does not facilitate easy opening
	204	Seal width less than 1/8 inch, but greater than 1/16 inch

1/ Any evidence of insect or rodent infestation shall be cause for rejection of the lot.

2/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause bag damage (for example, glass, metal filings, etc.) or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential bag damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one bag face (localized and aggregate).

c. Water spots.

3/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Bags exhibiting this type of delamination shall be tested by manually flexing the delaminated area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delamination area shall then be rapidly flexed by rotating both hands in alternating clockwise - counterclockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the bag material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, it shall be considered evidence of delamination.

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4/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide, from side seal to side seal that produces a hermetically sealed bag.

5/ To test for flex fractures the inside surface of the tray-shaped body shall be placed over a light source and the outside surface observed for the passage of light. Observation of light through the bag material in the form of a curved or straight line greater than 2mm in length shall be evidence of the presence of flex fractures in the aluminum foil.

6/ Applicable to form-fill-seal bags only.

7/ Applicable to preformed bags only.

4.5.5 Product examination. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE II. Product defects 1/ 2/

Category		Defect
Major	Minor	Applicable to dehydrated product
101		Presence of incomplete dehydration such as damp or soggy areas
102		Evidence of thawing during freeze drying as indicated by glazed or collapsed areas
103		Evidence of scorching or other mishandling during dehydration, as indicated by uncharacteristic color and/or flavor
104		Block is broken or crushed 3/
	201	Dimensions not as specified
	202	Inside bag dimensions not as specified 4/
		Applicable to rehydrated product 5/
105		Fruit does not rehydrate in specified time
106		Any piece of the rehydrated fruit shows incomplete penetration of water to the center

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TABLE II. Product defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	Applicable to rehydrated product 5/ (cont'd)
107		Product does not have characteristic color, odor, and flavor of the specified kind

1/ The presence of foreign material (for example, glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot (see 3.4.3).

3/ A broken block is one that is broken into five or more distinct pieces or which has more than 1/8 of its volume missing. A crushed block is one in which 1/8 or more of the volume of the block has been reduced to crumbs or powder.

4/ Applicable to form-fill-seal bags only.

5/ Rehydrate in 2 fluid ounces of cold water (55 $\pm$ 5°F) for the following time period.

a. Kind I or II - 20 minutes maximum
b. Kind III or IV - 15 minutes maximum

4.5.6 Bag vacuum examination. Not less than 96 hours after filling and sealing the bags shall be visually examined for conformance to the vacuum requirements in 5.1.1.1.2. The sealed bag shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.7 Moisture and sulfur dioxide content testing. The finished product shall be tested for moisture content and sulfur dioxide content as specified in 4.6.1 and 4.6.2 (kind IV product is not required to be tested for sulfur dioxide content). The lot size shall be expressed in bags. The sample unit

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shall be the contents of a sufficient number of bags to perform both tests. Any moisture content test result exceeding 2.5 percent or any sulfur dioxide content test result of less than 200 or more than 1100 parts per million shall be considered a defect. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 applied separately to each characteristic.

4.5.8 Bag closure seal strength testing. The filled and sealed bags shall be tested for closure seal strength in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply tensile load to the specimen by movement of the upper or lower clamp may be used. The test specimens shall be cut to a length suitable for proper mounting. For preformed bags, three adjacent specimens, 1/2 or 1 inch wide shall be cut from the closure seal of each bag in the sample. For the form-fill-seal bags, three adjacent specimens, 1/2 or 1 inch wide shall be cut from each side and each end of each bag in the sample. For the preformed bag, the average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from that seal. For the form-fill-seal bag, the average seal strength of each side and end of the bag shall be calculated by averaging the test results of the three specimens cut from that side or end. The results shall be reported to the nearest 0.1 pound per inch of width. Any test specimen or average seal strength failing to meet the requirements of 5.1.1.1.2 or 5.1.1.2.1 shall be classified as a major defect. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.9 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack, markings missing, incorrect, or illegible.
Cushioning material missing or not as specified.

Minor: Number of bags not as specified.
Bags not packed flat in layers.
Other required markings missing, incorrect, or illegible.

4.6 Methods of inspection

4.6.1 Moisture content test. Moisture determination shall be made by the vacuum oven method. Grind not less than 25 grams of a well-mixed sample in a clean, dry Waring Blendor or similar mill for 1 minute. Weigh approximately 2 grams of the ground material into a tared, dry, aluminum weighing dish

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(approximately 2 to 2-1/2 inches in diameter and 3/4 inch in depth with tight-fitting cover). Place the dish with cocked lid in a vacuum oven and dry for 16 hours at 60°C under a pressure of not more than 100 mm of mercury. During drying, admit to the oven a slow current of air (approximately two bubbles per second), dried by passing through concentrated sulfuric acid. Remove the dish and allow to cool in a desiccator before weighing. Calculate the percent of moisture in the original sample and report the result to the nearest 0.1 percent.

4.6.2 Sulfur dioxide content test. Sulfite (calculated as sulfur dioxide) shall be determined on the sample prepared as specified in 4.6.1 in accordance with the Total Sulfurous Acid method in the Chemical Preservatives Section of the Food Additives: Direct Chapter of the Official Methods of Analysis of the Association of Official Analytical Chemists. The results shall be reported to the nearest 10 parts per million.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. One block of one kind of product as listed below shall be unit packed in a barrier bag as specified in 5.1.1.1 or 5.1.1.2.

Kinds	Block weight (grams)
I	13.0
II	12.0
III	16.0
IV	14.0

5.1.1.1 Preformed bags. The preformed bags shall be fabricated from 0.002 inch thick isomer or polyethylene film laminated or extrusion coated to 0.00035 inch thick aluminum foil which is then laminated to 0.0005 inch thick polyester. The three plies shall be laminated with the polyester on the exterior of the bag. The complete exterior surface of the bag shall be uniformly colored in the range of green 34079 through 34087 or 24052 through 24087 or brown 30045 through 30118 (excluding 30109) or 10045 of FED-STD-595. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being unit packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into bags.

5.1.1.1.1 Preformed bag construction. The preformed bag shall be fabricated by heat sealing to the size and design configuration as shown in figure 1. Each bag shall be provided with an extended fold over lip and tear notches as shown in figure 1. The bag shall maintain its integrity and air tightness of the side and bottom seals when tested as specified in 4.5.6.

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The side and bottom seals shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch when tested as specified in 4.5.1.3.1.

5.1.1.1.2 Preformed bag filling and sealing. The fruit block shall be filled into the bags in a manner to avoid puncturing the bag material. The filled bag shall be closed under a vacuum of not less than 23 inches of mercury with a 3/8 ± 1/8 inch heat seal. If thermal impulse closure seals are used, a seal width of 1/8 inch will be acceptable. The closure seal shall be free of entrapped matter (fruit crumbs, moisture, etc.) that reduces the effective closure seal to less than 1/16 inch wide. The closure seal shall not extend below the tear notch on either side of the bag. The closure seal shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch (see 4.5.8). The sealed bag shall show no loss of vacuum when examined as specified in 4.5.6.

5.1.1.2 Form-fill-seal bags. The form-fill-seal bag shall consist of a formed tray-shaped body with a flat-sheet, heat sealable cover or a tray shaped body with a tray-shaped heat sealable cover. The tray-shaped body and the tray-shaped cover shall be fabricated from a 3-ply flexible laminate barrier material consisting of 0.0009 inch thick oriented polypropylene bonded to 0.0007 inch thick aluminum foil with 10 pounds per ream pigmented polyethylene and bonding the opposite side of the aluminum foil to 0.003 inch thick ionomer or a blend of not less than 50 percent linear, low density polyethylene and polyethylene. The linear low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1, having a melt index range of 0.8 to 1.2 6/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 G/CC in accordance with ASTM D 1505. Alternatively, 0.0005 inch thick polyester may be used in place of the oriented polypropylene as the outer ply of the laminate. The flat sheet cover shall be made from the same 3-ply laminate as specified for the tray-shaped body or tray-shaped cover, except that the aluminum foil thickness shall be 0.0005 inch. The color requirements of the exterior (oriented polypropylene or polyester side) of the laminates shall be as specified in 5.1.1.1. The laminates shall be suitably formulated for food packaging and shall not impart any odor or flavor to the product being unit packed.

5.1.1.2.1 Form-fill-seal bag construction. The tray-shaped body and the tray-shaped cover shall be formed by drawing the flexible laminate material into an appropriately shaped cavity. The flat cover shall be in the form of a flat sheet of the barrier material taken from roll-stock. One fruit block shall be placed into the tray-shaped body of the bag. The filled bag body shall then be hermetically sealed in accordance with the vacuum requirements as specified in 5.1.1.1.2. Bag closure shall be effected by heat sealing the flat cover or the tray-shaped cover to the tray-shaped body along the entire bag perimeter. The inside bag width and length shall be not more than 4-3/4

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and 6-1/8 inches, respectively. The closure seal width shall be not less than 1/4 inch wide. The closure seal shall be free of entrapped matter (fruit crumbs, moisture, etc.) that reduces the effective closure seal to less than 1/16 inch wide. The closure seal shall have an average seal strength of not less than 6 pounds per inch of width and no individual specimen shall have a seal strength of less than 5 pounds per inch (see 4.5.8). One or two tear notches in accordance with the tear-notch details in figure 1 shall be provided on the bag outside edge to facilitate tear opening in the machine direction of the bag laminate material. A tear-notch shall be placed on one outside edge or two opposite outside edges of the bag. Each tear-notch shall be located not more than 1-1/4 inches or less than 1/4 inch from an outside corner of the bag. Alternatively, if the bag has serrated edges, the serrations may be used as tear-notches provided that the serrations are located to effect easy opening in the machine direction of the bag laminates, the serrations are sharp (no plastic tailing exists), and the serration depth and the minumum seal width at the serrations are in accordance with the notch details in figure 1. The sealed bag shall not show any evidence of material degradation, aluminum flex fractures, delamination or foreign odor when heat sealed into bags.

5.2 Packing. Packing shall be level C.

5.2.1 Level C, for shipment to ration assembler. Not more than three hundred and twenty-five unit packs of one kind of product shall be packed flat in layers in a snug-fitting fiberboard box constructed and closed in accordance with style RSC-L, type CF, class domestic, variety SW, grade 175, of PPP-B-636. Cushioning material not less than 1/8 inch thick, in accordance with type II, class B of PPP-C-843 or cushioning material not less than 1/16 inch thick in accordance with PPP-C-1797 or with type VII, class 1, PPP-C-1752 shall be placed between each of the layers and at the bottom and top of the box to minimize movement of the individual unit packs.

5.3 Labeling and marking.

5.3.1 Unit packs. Each unit pack (bag) shall be clearly printed with permanent ink in large, clear black letters:

FRUIT MIX
PEACHES (As applicable)
PEARS
STRAWBERRIES

EAT DRY OR RECONSTITUTE IN 2 OUNCES COLD WATER

TO RECONSTITUTE: COVER FRUIT WITH 2 OUNCES WATER, LET SOAK UNTIL SOFT

NAME AND ADDRESS OF CONTRACTOR

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5.3.2 Shipping container. Shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Kind of product required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.4.3 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Fruit pressure tester. It has been found that McCormick Fruit Pressure Tester, Model FT 011, equipped with a 5/16 inch plunger, supplied by McCormick Fruit Tech, has been found to perform satisfactorily in the determination for hardness of pears (see 3.2.3).

6.4 Subject term (key word) listing.

Dried
Flexible bag
Individual portion MRE

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8915-0852)

Review activities:

Army - MD, QM
Navy - MC
DP - SS

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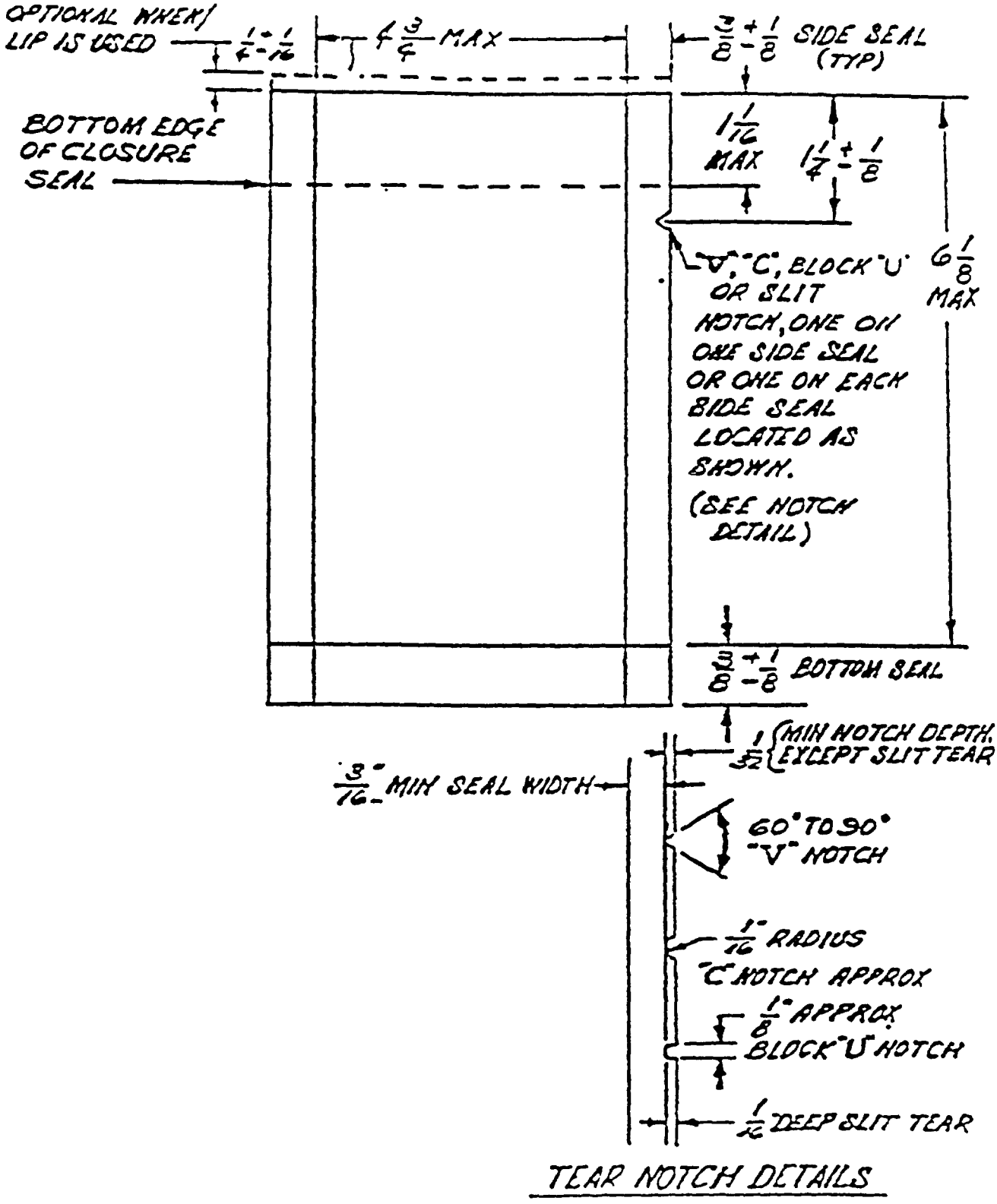


Figure 1 - Preformed Bag

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given
2. The submitter of this form must complete blocks 4, 5, 6, and 7
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-44070C	1991 April 24

3. DOCUMENT TITLE

FRUITS, FREEZE-DEHYDRATED, FOR MEAL, READY-TO-EAT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON 256-4501
c. ADDRESS (Include Zip Code) Commander, U S. Army Natick RD&E Center ATTN: STRNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT. Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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